



The counterfactual challenge: How machine learning can enhance policy evaluation

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Received 17 February 2025; Received in revised form 19 April 2025; Accepted 12 May 2025
Available online 12 July 2025

Abstract

Understanding the impact of public policies and programmed actions is a complex and challenging task that deserves attention and resources. Even if the economic literature on counterfactual methods and identification is well established, conventional methodologies still struggles with holding up the *unconfoundedness assumption* in complex socio-economic and policy contexts. This paper sets up a ‘counterfactual challenge’, testing the ability of conventional matching versus a novel application of Supervised Machine Learning classification process to identify suitable counterfactuals. Working with high-dimensional data in complex socio-economic and policy contexts, results show that Machine Learning algorithms are better equipped to effectively balance treatment and control groups across a wide range of covariates compared to conventional matching methods. In the context of decision making and policy planning, we show the potential of Machine Learning to drastically improve the reliability and precision of information supporting policymakers in their choices. This is argued to have a positive impact on the effective use of public resources especially in complex and underdeveloped areas and contexts. Improvements in the precision of impact evaluation could result in significant gains in resource efficiency, both by generating realistic expectations of policy outputs, and improving scarce resource allocation we show that the use of Machine Learning algorithms for counterfactual identification consistently provides more precise results and supports policymakers in navigating complex contexts.

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Keywords: Counterfactual Identification; Machine Learning; Policy impact; EU policy; Developing regions

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1. Introduction

Evaluating the impact of policies and public programmes is a crucial process that aids policy-makers in decision-making and program adjustments. This task, though complex and challenging, can yield valuable insights, particularly in intricate policy and socio-economic contexts.

Indeed, estimation of the impact of past policies, simulation on future outputs, and the overall knowledge of the context under investigation are today's cornerstones of policymaking; their reliability and precision are of paramount importance and can often lead to a better use of scarce public resources.

Over the past decades, the econometric literature on estimating *ex post* causal effects has seen significant growth both in terms of applications and of methodologies implemented. In detail, the econometrics of policy evaluation has developed various methods and techniques, leveraging both the increasing computational power and the availability of new, more detailed data. Despite the significant technical growth, however, the general framework, established by the Rubin Causal Model (Imbens & Rubin, 2010) in the early 1990s, remains unchanged. It aims to establish a causal link between the policy (treatment) and the observed impact on treated units, compared to the hypothetical outcome if the same units had not received the treatment (Athey and Imbens, 2017). Unfortunately, such a hypothetical outcome is not directly observable (in our example, units have, indeed, been treated) and we need to rely on the observable outcome of other (untreated) similar units. The estimation of a proxy for this untreated group is conceptually simple but difficult to obtain in practice. In theory, this could be easily estimated, if the treatment is assigned randomly. However, random experiments are extremely rare in social sciences where treatment (i.e., a public policy) is not assigned randomly but with the explicit purpose of addressing known issues. In policymaking, the selection of treated units (where or at whom to address specific policy) and the expected outcome of the policy (the issue the policy addresses) are organically interconnected.

Scholars have developed several methods and techniques to deal with this problem identified as the *unconfoundedness assumption*. This assumption requires all confounding factors (i.e., factors correlated with both potential outcomes and treatment assignments) to be observed and controlled. Under this assumption, as shown by Rosenbaum and Rubin (1983), the treatment can be considered as being randomly assigned, and we can give a causal interpretation to the average difference between treated and untreated units.

This area of study, often referred to as counterfactual methods and identification, is mature, and its competing estimators have found various applications over time (Athey and Imbens, 2017; Quillien, Lucas, 2023; Guidotti, 2022). However, despite its maturity, the challenge of identifying suitable counterfactuals and adhering to the *unconfoundedness assumption* still remains. While the growing availability of data allows for the observation and inclusion of a larger set of confounding factors in the analysis, commonly used techniques have not evolved at the same pace.

This paper presents a 'counterfactual challenge', testing the ability of different methodologies to identify suitable counterfactuals working with high-dimensional data in complex socio-economic and policy contexts. By leveraging both these elements (large availability of data and complex contexts where policy output and treatment selection share several factors and dynamics) the challenge stresses to a maximum the ability of different methodologies to identify suitable counterfactuals.

In our specific case, this is done using real-world data on firms participating in environmental and carbon reduction programs, and with extensive balance sheet information available. In the challenge, the paper aims to compare the performance of the most commonly used and regarded technique, the propensity score matching method (Becker & Ichino, 2002) using a linear

parametric algorithm to identify similar untreated observations and estimate probability weights, with the performance of Machine Learning algorithm tasked to perform the same operation. Of course, while in this paper we focus on this specific case, the implication of our challenge extends to all the situations where commonly used techniques of policy evaluation are replaced by Machine Learning algorithms.

In recent years, task-oriented Supervised Machine Learning methods have emerged as a viable alternative to propensity score matching for estimating probability weights, thanks to their ability to control for a large number of covariates (Athey and Imbens, 2017). The field of economics, especially the literature related to policy impact evaluation, is lagging behind other fields in the use of Machine Learning methods, especially for task-specific applications that are already addressed using other methods. While there has been a recent proliferation of the use of some types of Machine Learning in the field of economics (see, for a review, De Lombaerde et al., 2023), the use of Supervised Machine Learning methods for the identification of counterfactuals is not among those.

From the methodological literature, and the application in different fields (i.e., Lee et al., 2010; Linden & Yarnold, 2017), we anticipate that the enhanced flexibility of Machine Learning algorithms and their expanded ability to handle a large number of covariates will yield more accurate and robust results.

This will allow for a stronger interpretation of the *unconfoundedness assumption* and the identification of more suitable counterfactuals.

While this challenge may initially seem to be of interest primarily to technical and methodological literature, its implications are far-reaching. Decisions made by policymakers, which impact entire territories and policy programs, are based on the insights and expectations generated by impact evaluation studies and reports. Rigorous and precise impact evaluation enables efficient use of public resources, mitigating issues arising from the scarcity of public funding, especially in complex and underdeveloped areas and contexts. Even a marginal improvement in the precision of impact evaluation could result in significant gains in resource efficiency, both by generating realistic expectations of policy outputs and improving scarce resource allocation.

The analyses and results presented in this paper contribute to the literature on policy impact evaluation and policy learning (Cerulli, 2021; Estrada et al., 2023; Newman & Mintrom, 2023). They align with the ongoing discussion on data-driven government learning (Castañeda and Guerrero, 2019) and the production of information to support decision-making.

The remainder of the paper is organized as follows: Section 2 (Background) introduces the policy context for the ‘counterfactual challenge’ and details the emergence of Machine Learning methods for policy impact evaluation. Section 3 (Materials and Methods) describes the real-world data used, and their sources, and provides a detailed specification of the two algorithms involved in the challenge. Section 4 (Results and Discussion) presents the main findings of the paper, comparing how different specifications perform in terms of counterfactual identification and estimated impact. Finally, Section 5 (Conclusions and Policy Implications) summarizes the paper’s overall contribution and discusses the potential impact of broader adoption of Machine Learning algorithms for counterfactual identification.

2. Background

2.1. The relevance and complexity of environmental and carbon reduction policy setting

Policymaking and decision-making are challenging and complex tasks to navigate where precise and reliable information and expectation can make the difference. Providing this reliable

information can be a complex operation even when the aims of the policy program are overall quite straightforward and devoted to addressing societal and global problems easy to pinpoint and understand.

This is the case of environmental and carbon reduction policies in Europe. What is the impact of these policies? What may seem like an obvious question reveals, in fact, quite a complex setting. Within the EU's public policy sphere, this simple research question can be reframed as follows: does the technology-driven initiative to support and encourage SMEs to invest in activities that optimize the use or reuse of water, energy, and materials effectively reduce our environmental impact? The EU initiative is, primarily, a technology-led effort that requires the active participation of economic actors for its full implementation. It can only work if a large part of firms and businesses operating in Europe buy into this vision. Taking at face value the potential environmental benefits of optimizing the use of energy, water, and resources in production activities, the effectiveness of the strategy hinges on the widespread of enabling technologies and practices which is, in turn, mediated by the economic and competitive benefits they could bring to economic actors (Verhoef & Nijkamp, 2003).

In the end, the question could be reframed as a micro-economic problem: does the adoption of these technologies provide firms with a competitive advantage and produce economic benefits for the actors involved? Despite the decade has only begun, answering this question will play an increasingly important role in the effectiveness of European carbon reduction and cohesion programs, helping to understand the potential of the strategy and, possibly, making adjustments. *Ex-post* evaluation of the current programs implemented all around Europe, and their adoption by micro-economic actors, are not possible yet (the 2021–2027 programming period is in its starting phase), but similar programs in previous EU programming periods (i.e., carbon reduction programs enacted in the 2013–2020 programming period) can be studied to understand the impact of their adoption and leverage information for policymakers to make adjustments to currently running programs.

With an expanding body of literature on the adoption of green transition and carbon reduction technologies (among others Clementi & Garofalo, 2023), there is no real consensus on the impact of these innovations when adopted by micro-economic actors. The environmental benefits of optimizing the use of water, energy, and materials are evident (Jaffe and Stavins, 1994; Miara et al., 2014); however, the economic benefits for microeconomic actors have only recently begun to be explored. This is a key element for environmental policies, especially in the EU context where economic actors' participation and adoption play a pivotal role.

Some studies suggest an overall increase of production costs associated with the implementations of carbon reduction innovations (Antonioli et al., 2022), while others individuate a causal link between public financial support and an improvement in material productivity and operating costs for certain industrial sectors (Flachenecker and Kornejew, 2019; García-Quevedo and Jové-Llopis, 2021) or with specific firm characteristics and managerial capacities (Leoncini et. al., 2019). Additionally, regional and territorial factors may play a role in the adoption and significantly influence these effects (Cainelli et. all., 2015), as well as the varying characteristics of microeconomic actors (Bassi and Dias, 2020; Leoncini et. al., 2019).

In summary, current literature does not provide a clear answer regarding the general expectations in terms of the economic benefits of adopting these technologies but suggests that such an adoption can relate to a varying set of firms' characteristics.

Reducing the complexity, and isolating the potential impacts from heterogeneous dynamics, can be quite difficult in these scenarios where adopters' self-selection is pronounced and influenced by several characteristics. Here, dealing with the *unconfoundedness assumption* and

provide robust estimates is extremely important to provide policymakers with reliable information.

Both the complexity, and the relevance, of this policy setting are enhanced by territorial socio-economic condition (Cainelli et al., 2015; Fanti et al., 2023) where in the lagging behind European regions the effective, and efficient, use of public resources - based on reliable information and expectations - is a central necessity.

All in all, the task of evaluating the micro-economic impact of environmental and carbon reduction policy in confined - lagging territories such as the Southern part of Italy (Fanti et al., 2023) - makes for a compelling, and complex, case study to test the ability of different methodologies in identifying counterfactual and hold up the *unconfoundedness assumption*.

2.2. Matching methods and the emergence of Machine Learning for policy impact evaluation

Since the early 2000s, economists have utilized various techniques to identify counterfactuals and ensure the comparison of treated units with similar but untreated ones. This involves generating observational weights for model input, reducing the control group to a subset of the available observations, or employing a combination of both strategies (Becker & Ichino, 2002).

While propensity score matching is one of the first implementations of the Rubin Causal Model (Imbens & Rubin, 2010) it is still considered nowadays among the most solid methodologies to use (Athey and Imbens, 2017). Heuristically, the generation of observational weights is commonly achieved using a different array of matching algorithms (Becker & Ichino, 2002), based on the use of binomial logistic regressions.

Binomial logistic regressions are quite simple and powerful; regressing on a defined set of covariates, they calculate the probabilities of each observation to be part (or not) of a specific group. Matching algorithms use these coefficients to identify similar but untreated observations and then generate observational weights. While this matching procedure helps addressing the issues raised by the *unconfoundedness assumption*, it can struggle to balance groups (treated and untreated) across large sets of characteristics (covariates), leading to increased risks of underfitting or overfitting when matching on a large set of covariates (Muller & Guido, 2017).

While this is not inherently problematic, it often forces researchers to relax this assumption when there are large sets of (observed) confounding factors and the algorithm is not able to balance them all.

In recent years the use of Machine Learning algorithms has emerged, in other disciplines, as a possible alternative to matching algorithms (i.e., Lee et al., 2010; Linden & Yarnold, 2017). The principal advantage of Machine Learning algorithms, particularly those from the Classification And Regression Tree (CART) family, consists in the capacity to handle and fit a large dataset and numerous characteristics (covariates) into models more effectively. The higher dimensionality on which nonparametric Machine Learning algorithms can operate, compared to commonly employed matching methods, has the potential to help researchers control for several different confounding factors altogether and identify more appropriate counterfactuals. A change in methodological perspective towards the use of Machine Learning processes and algorithms, instead of matching ones, will allow for a stricter interpretation of the *unconfoundedness assumption* and, in turn, produce more precise and reliable estimation (Athey & Imbens, 2019). Precision and reliability are key elements for the efficiency of decision-making and the effectiveness of policy programs (Mergoni & De Witte, 2022); complex socio-economic and policy settings, such as the one presented in this paper could benefit the most from these

improvements. Indeed, the potential improvement in the reliability of estimates can support policymakers in making better-informed decisions and planning policy adjustments for future implementations.

3. Model and materials

Two different methodologies are pitted against each other in what we call a 'counterfactual challenge'. Both a Propensity Score Matching and a Supervised Machine Learning classifier will be tasked to identify counterfactuals and estimate Inverse Probability Weights (IPWs). The selection of these counterfactuals is a crucial part in the estimation of the impact of public policy, which is performed with a Difference-in-Difference (DiD) framework as detailed below. In the next [Section \(4. Results and Discussion\)](#), the results of these estimations will be tested in their capacity of dealing with high-dimensionality and confounding factors in order to hold up to the *unconfoundedness assumption*.

The treatment group consists of firms that have implemented activities to optimize the use or reuse of water, energy, and materials in 2016¹ with investments from the EU's Cohesion Policy fund limited to the Southern part of Italy (to exclude heterogeneity due to macro-economic dynamics). For the study, a large panel dataset of firm-level yearly balance sheet information (AIDA, Bureau Van Dijk) on firms operating in developing regions in Italy² for the period 2009–2019 has been collected.

The dataset includes a wide range of variables describing different characteristics of firms in the pre-treatment period and includes all available information identified as possibly relevant from the literature ([Antonioni et al., 2022](#); [Clementi & Garofalo, 2023](#); [Flachenecker and Kornejew, 2019](#); [García-Quevedo and Jové-Llopis, 2021](#); [Leoncini et. al., 2019](#)). The selected covariates are described in Table A1 in Annex. Treated and untreated firms exhibit very different characteristics, especially in terms of firms' size (i.e., employment, total tangible assets) and age. Moreover, the treatment group is also quite unbalanced in terms of sample size, comprising only around 300 out of 20k total observations.

Following methodological and empirical literature (in fields different from economics) on the use of Supervised Machine Learning classifiers for counterfactual identification ([Athey and Imbens, 2017](#); [Lee et al., 2010](#); [Linden & Yarnold, 2017](#)), we select a Random Forest algorithm classifier - an ensemble of Decision Trees Classifier ([Breiman, 2001](#)) - to estimate Inverse Probability Weights (IPWs) and identify neighbouring observations in the untreated.

The Random Forest algorithm is trained to classify treated and untreated observations on a random subset of the observations. Tested over unseen data, it correctly identifies 95 % of the treated observations (recall of 5 %) with an overall accuracy of 98.01 %. High accuracy indicates the reliability of the classifier, which can competently decide whether an observation is treated or not based on the pre-treatment features provided. The trained algorithm is run on the observational data, assigning a specific probability of being treated or not to each observation. Following [Lee et al., \(2010\)](#) and [Linden & Yarnold, \(2017\)](#), these probabilities are treated as

¹ A single year in the programming period has been chosen to avoid heterogeneity due to different times of implementation. 2016 has been selected for being in the middle of the programming period when most of the implementations happen. Firms participating in the same programs, but with implementations in different years have been excluded from the analysis.

² These regions are: Basilicata, Calabria, Campania, Molise, Sicilia, Puglia and Sardegna.

IPWs in the DiD estimation, assigning greater weight to observations that the algorithm believes are or should be treated.

Alongside these probabilities, the Random Forest creates a proximity matrix between each observation. This is exploited to identify the five most similar untreated observations for each treated unit, without replacement. This is done to reduce the analysis to smaller, but more similar, counterfactuals which is a commonly employed technique in matching methods.

The commonly employed methodology, pitted against the Random Forest classifier in this challenge, is the Propensity Score Matching method (Becker & Ichino, 2002). The propensity Score Matching is the most used, and regarded, implementation of the Rubin Causal Model (Athey and Imbens, 2017). It generates IPWs using a linear parametric model, a logit algorithm, and can leverage Nearest Neighbour algorithms to identify subgroups of - most similar - untreated observations.

Both methodologies are able to perform the same tasks, estimate IPWs and identify neighbouring observations. For the 'counterfactual challenge' different specifications have been delineated to give both methodologies their best chances: both methodologies are run over the full set of available covariates estimating IPWs for all untreated observations and IPWs only for neighbouring observations (for each treated unit we tasked the algorithms to identify five untreated units). Moreover, the Propensity Score Matching is also run on a reduced set of covariates for which we are sure the algorithm is able to balance the (limited) characteristics between treated and untreated groups.

All results are presented for comparison.

The specifications' adherence to the *unconfoundedness assumption*, i.e. the algorithm's capacity to balance confounding factors and identify suitable counterfactuals are tested on both individual covariate balance and the overall goodness of fit with a range of statistical tests detailed in the next section.

DiD estimations, for the growth in Operating Margins at the firm level, are performed for each specification as in equation 1 below. The estimation is performed for all those specifications which abide to the DiD model's assumption (the *parallel trend assumption*), regardless of the questionable fit and covariate balancing of the specific counterfactual group.

$$\Delta DID_{PSM} = \frac{1}{N_1} \sum_{i \in N_1} \left[Y_{it} - \sum_{j \in N_0} w_{ij} Y_{jt} \right] - \frac{1}{N_1} \sum_{i \in N_1} \left[Y_{it-1} - \sum_{j \in N_0} w_{ij} Y_{jt-1} \right] \quad (1)$$

where:

Y : output for units i and j at time t ,

N_1 : treated units,

N_0 : control units,

w_{ij} : weight for control unit j when matching with treated unit i , calculated as

$$w_{ij} = K(e(X_i, Z_i)e(X_j, Z_j))$$

Z_{ij} : firm-level pre-treatment covariates.

In the short-term, following the implementation of green technologies and practices, the growth in Operating Margins should provide a direct metric for potential efficiency gains in energy and material usage. A counterfactual design, which not only respects the parallel trend assumption but also demonstrates that the comparison is conducted on a similar group, enables the establishment of a causal link between the implementation of these practices and technologies and the output.

Table 1
Balancing and difference in difference estimation.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Full covariates set		Reduced covariates set		Full covariates set	
	Baseline	LOGIT IPW	LOGIT IPW n5	LOGIT IPW	LOGIT IPW n5	CART IPW	CART IPW 5n
Pre-Treatment Covariate Balancing (reported standardized difference)							
Operating Margins	0.339	0.316	0.217	0.268	0.162	0.155	0.107
Age	0.279	0.234	0.142	0.323	0.049	0.038	0.016
N. of Employees	0.408	0.801	0.382	0.827	0.201	0.096	0.016
Value Added	0.446	0.482	0.281	.	.	0.084	0.005
personnel Cost	0.397	0.397	0.278	.	.	0.055	0.032
Revenues from Sales	0.450	0.160	0.054	.	.	0.049	0.038
Tangible Assets	0.248	0.068	0.166	0.039	0.194	0.018	0.054
Reliance on Tangible Assets	0.637	0.502	0.677	.	.	0.056	0.016
Return On Investments	0.083	0.102	0.156	.	.	0.024	0.038
Net Profits	0.200	0.249	0.005	.	.	0.011	0.023
Debt to Ebitda Ratio	0.194	0.317	0.133	.	.	0.044	0.037
Nace Sectors	YES	YES	YES	YES	YES	YES	YES
NUTS–2 Region	YES	YES	YES	YES	YES	YES	YES
Model’s overall goodness of fit							
Rubin’s B	56.4^	16.6	45.5^	22.4	31.1^	21.9	19.8
Rubin’s R	2.34^	0.42^	0.09^	0.82	0.06^	0.77	0.50
Difference in Difference coefficients							
Parallel Trend before Treatment	NO	NO	NO	YES	YES	YES	YES
DiD - Operating Margins	.	.	.	.279**	.139***	.157***	.140**
Obs	21200	21200	1795	21200	1772	21200	1920
Treated	320	320	320	320	320	320	320
R2	0.5	0.798	0.771	0.674	0.756	0.696	0.696

*** $p < .01$; ** $p < .05$; * $p < .1$
^ if B > 25 %, R outside [0.5;2]

4. Results and discussions

The main results are presented in Table 1, which provides a comparison of the balancing and DiD estimation across different models’ specifications. The first column (1) serves as a reference point, operating on all available observations without IPWs.

The upper section of the table displays the ‘between groups’ balance estimated with an F-test for each covariate independently, followed by the overall goodness of fit via Rubin’s B and Rubin’s R (Rubin, 2001).³ These, together inform the adherence to the *unconfoundedness*

³ These are common metrics employed to evaluate the overall fit between a treated and untreated group.
Rubin’s B measures the difference in means over standard deviation of the IPWs between the treated and untreated groups, an estimation under 25 % is considered balanced. Rubin’s R measures, instead, the ratio of the variance of the IPWs in the treated group to the variance of the IPWs in the matched sample; and estimated value between 0.5 and 2 indicates a good fit.

assumption of each specification. Specifications that can better balance a larger number of covariates and show stronger goodness of fit are implicitly better with confounding factors, thus identifying better suiting counterfactuals.

The lower section of the table reports the estimated DiD coefficients and indicates whether evidence of parallel trend was found.

When applied to real-world high-dimensionality observational data from developing regions in Italy, the Random Forest (CART) algorithm noticeably outperforms logit matching in identifying similar observations and dealing with the *unconfoundedness assumption* by weighting observations based on their probability of being treated.

The top part of the table reports the standardized difference for each covariate; these metrics are used to define the levels of imbalance of a continuous variable between two groups. While there is no empirical evidence, or mathematical reasons to support the use of a specific cut-point, several authors use different cut-offs. Rubin (2001) suggests a cut-off of 0.25 alongside Cohen (1988) who argues for a similar cut-off of 0.20 signalling a "small" enough effect. The most rigid cut-point is identified in Normand et al. (2001), where is suggested that a standardized difference greater than 0.10 is indicative of imbalance. While the discussion on the optimal cut-off to individuate imbalance between groups is not the focus of this contribution, it emerges that standardized differences as close to zero as possible should be the aim for a balanced comparison (Linden & Samuels, 2013).

The balance of covariates is consistently superior in CART specifications (columns 6 and 7), with only one of them being on the edge of the 0.10 cut-point individuated by Normand et al. (2001). In comparison, most covariates in the logit matchings (columns 2 and 3 and columns 4 and 5) are not able to stay within the suggested cut-off of 0.20 or even 0.25. Furthermore, the CART specifications' overall goodness of fit is significantly stronger compared to logit specifications, whether the latter is run on the full or reduced set of covariates designed for the algorithm to be able to best balance. Notably, Rubin's B is always under 25 % in CART specifications, which is not the case for all logit specifications. Additionally, Rubin's R, which indicates the ratio of Inverse Probability Weights' (IPWs) variance, remains within acceptable boundaries only in CART specifications, suggesting a lower risk of overfitting and bias introduction. Implementing a reduction of the counterfactual group to the five most similar observations (columns 3, 5, and 7) shows improvements in covariate balancing across all specifications. However, only the CART model in column 7 maintains a robust goodness of fit. The Random Forest uniquely identifies the five most similar observations for all 320 treated units, totalling 1600 untreated observations. In contrast, matching algorithms could not identify five similar untreated observations for each treated unit, further reducing the counterfactual group to 1475 and 1452 observations.

Effectively, using the Random Forest (CART) algorithm instead of the more common logit matching to identify similar observations and develop weights, we are better able to deal with the *unconfoundedness assumption* and it allows for a more precise and rigorous comparison. This is important in the measure that, only in a setting of rigorous and balanced comparison statistically significant, results can aim to explain the causal relationship in question. Indeed, following Rubin Causal Model (Imbens & Rubin, 2010), treatments can be considered as assigned like-random (and the causal relationship established) only if confounding factors are observed and *unconfoundedness assumption* is respected.

The DiD analysis, presented in the lower part of Table 1, confirms the better fit of the counterfactuals identified via CART algorithms. Indeed, matching specifications on the full set of covariates failed to identify counterfactual groups exhibiting either a strong balance or evidence of a parallel trend before the intervention (columns 2 and 3). Unreliable results, from these specifications, were not reported. DiD coefficients were estimated and reported only via matching algorithm with a reduced set

of characteristics and via Random Forest algorithm (CART). Both methods yield positive and significant results, but the intensity of the estimated DiD coefficients varies. In comparing specifications, it is important to keep in mind the balanced comparison (*unconfoundedness assumption*) and parallel trend assumptions of the difference in difference model validating the results. The specification in column 5 (logit matching algorithm on the five most similar observations on a reduced set of covariates) fails to pass tests for the overall goodness of fit, with both metrics suggesting the model introduces a strong bias to the estimates.

CART specifications, working with a larger set of covariates, hold to a more rigid interpretation of the *unconfoundedness assumption*. The balancing of each covariate is markedly better in CART specifications, and they are able to do so without introducing problematic levels of bias. Reducing the counterfactual group to only the five most similar observations trade a marginal improvement in covariate balancing for a larger bias (a lower variance in the IPWs, sitting on the lower bound of the Rubin's R).

Overall, the CART specifications are more credible due to a superior goodness of fit and balance across a larger set of covariates. In comparing the results and estimated coefficients, it is noticeable how the two CART specifications are also more similar between them and show the smallest difference.

All specifications able to support a causal interpretation of the results (columns 4, 6, and 7) indicate a short-term increase in production efficiency for the treated firms, but the increase estimated using CART specifications is substantially smaller. This suggests that relying on common matching algorithms, balancing treated and untreated over a smaller set of covariates (column 4) with a relaxed *unconfoundedness assumption*, could lead to a misleading estimation of the policy and the impact of firm-level implementation of publicly financed green technologies and practices. Such miss-estimation likely arises from the complexity of the comparison due to the unique characteristics of firms participating in these programs. Table 1 shows that, when employing traditional matching algorithms - even on a reduced set of characteristics - the researcher must choose whether to prioritize either a more balanced comparison (i.e., specification 5) or a lower introduction of bias in the analysis (i.e., specification 4). These choices can drastically change the outputs of the analysis. CART algorithms are better equipped to handle the complexity of these settings and produce more robust and consistent results.

5. Conclusion and policy implications

Dealing with the complexity of understanding the micro-economic impact of carbon reduction policies, this paper explores, to the best of our knowledge, for the first time in literature, the application of task-oriented Supervised Machine Learning, specifically algorithms from the CART family, for counterfactual identification.

The paper sets up a 'counterfactual challenge' where Propensity Score Matching methods and Supervised Machine Learning methods are all implemented on the same high dimensionality real-world data tasked to identify counterfactuals and estimate Inverse Probability Weights (IPWs) that are used to evaluate the impact of carbon reduction policy in the Southern part of Italy. The real-world case study is exploited, due to its innate complex setting and the large number of confounding factors, to test whether the non-parametric CART algorithm can be better suited to deal with the all-important *unconfoundedness assumption* compared to commonly employed matching methods.

Being able to identify a suitable counterfactual is a key element to understanding the economic impact on adopters and establishing a causal link with the implementation of specific technologies and practices. Complex scenarios such as the one presented in this paper, often

show conflicting evidence. We argue that this can be, at least partially, due to comparison problems and confounding factors between adopters and other micro-economic actors. Thanks to the advances made in the last decades, both in terms of data production and computational capacity, today the *unconfoundedness assumption* problem (Athey and Imbens, 2017) can be addressed with novel frameworks and methodologies.

The study serves as an exemplary case, fully demonstrating the potential of task-oriented Supervised Machine Learning algorithms in economics and their competitive advantage over commonly employed matching techniques in identifying counterfactuals and reducing complexity due to confounding factors.

The comparison of various counterfactual specifications, generated through different algorithms, confirms that the identification of a counterfactual control group and its Inverse Probability Weights (IPWs) identified using a Random Forest (CART) algorithm is significantly better than those from a standard logit matching specification. This observation aligns with expectations drawn from methodological literature in other fields and studies performed on simulated data, suggesting that CART algorithms are markedly better at identifying counterfactuals, especially when dealing with a large array of covariates, and produce more precise, reliable, and consistent results allowing for more rigid interpretations of the *unconfoundedness assumption*.

From an economic perspective, particularly from the perspective of evaluating the firm-level impact of implementing publicly subsidized green and carbon reduction technologies, the paper underscores the importance of meticulous and thorough counterfactual identification to establish causal links regarding the effects of policies and interventions. In our specific case, we have identified and discussed the short-term increase in production efficiency for treated firms, and we have stressed how consistent and reliable our results are, thanks to the novel approach proposed.

The results of our challenge can easily be extended. In scenarios such as environmental and carbon reduction policies, or more broadly, in cases with significant adopters' self-selection and several confounding factors, Supervised Machine Learning algorithms like the implemented Random Forest are better equipped to effectively balance treatment and control groups across a wide range of covariates. The resulting counterfactuals ensure a higher quality of comparison. At the very least, the outcomes of this study advocate for expanded exploration of Supervised Machine Learning techniques for handling large datasets to identify suitable counterfactuals and deal with the *unconfoundedness assumption*, supporting with clear evidence the burgeoning academic discourse on the impact of environmental and carbon reduction policies.

Findings from the comparison of specifications indicate that the short-term effects of these policies on firms may have been previously miss-estimated. Indeed, conventional matching techniques are limited in their ability to balance treatment and counterfactual groups and meeting models assumptions. Table 1 has shown how the results they produce could be sensibly different whether we focus on meeting the model assumption (i.e., parallel trend, balanced comparison) or including a larger set of characteristics in the identification of the counterfactual itself. Non-parametric Machine Learning algorithms help bridging this gap and provide solid results (Athey & Imbens, 2019).

We believe that our contribution constitutes a big step forward for policy learning which could be especially relevant for lagging and less endowed territories such as (but not limited to) the Southern part of Italy (Fanti et al., 2023), where understanding the ex-post impact and the future potential of policy programs by addressing the complexity of the setting is a key element for the effective use of public resources to produce meaningful impacts and competitive advantages. Task-oriented Supervised Machine Learning based counterfactual identification help to better inform policymakers in adjusting and tailoring policy programs to these contexts and allows for a more effective - and efficient - decision-making and use of public resources.

CRedit authorship contribution statement

Federico Fantechi: Conceptualization, Methodology, Formal Analysis, Data Curation, Drafting the manuscript, Writing; Alessandro Cusimano: Conceptualization, Writing.

Acknowledgments

Authors disclose that this study was funded by the European Union - NextGenerationEU, in the framework of the GRINS - Growing Resilient, INclusive and Sustainable project (GRINS PE00000018 - CUP B73C22001260006).

Annex

Table A1
Description of all control variables included in the balancing with mean values for treated and untreated for the period before the treatment

Variable	Description	Treated	Untreated
Age	Number of years passed from the reported year of incorporation of the firm.	25	22
N. of Employees	Number of Employees reported by year.	37.4	11.2
Value-Added	Value-Added/Employee reported by year.	1820.3	491.8
Revenues from Sales	Income generated from the sale of goods and services, by year.	8790.3	2387.6
Reliance on T.Assets	Share of Tangible assets over Total Assets reported by year.	0.10	0.14
Net Profit	Net earnings generated after taxes and operation expenses, by year.	204.3	12.1
Debt to Ebitda	Share of earnings available to pay down debt before tax, interest, amortization, and depreciation, by year.	4.2	2.7
Nace Sector	2-Digit Nace Rev.2 Industrial Sector.	.	.
NUTS–2 Region	NUTS–2 Region where each firm operates.	.	.
Dependent Variable			
Operating Margins	Profit generated after paying costs of production, by year.	417.8	73.6

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